Long / Short / Flip — Two-Layer Directional Spec for TradePilot

Source: deep-research run 2026-06-08 (5 angles, 23 sources, 101 claims → 25 verified → 16 confirmed / 9 killed by 3-vote adversarial check). Cited but practical.

Why this exists

Today (BEAR, Nifty -0.94%) v4 (long-only) lost -Rs8.3k while v5 (short-tilted) made +Rs465. The audit traced the entire loss to one class — `LONG_IN_BEAR` (78 trades, -Rs18.8k). Root cause: the scorer emits **0 SELL** signals, so the engine has no sanctioned way to be short and no gate to stop longing a falling market. This spec is the fix: a **regime gate** that decides the *allowed side*, plus an **intraday state machine** that flips within it.

Layer 1 — Daily allowed-side regime (the GATE)

Computed once per day per stock on **daily** bars. Output: `allowed_side ∈ {LONG_ONLY, SHORT_ONLY, BOTH, FLAT}`.

Step	Rule	Data (have it?)
Trend permission	`ADX(14) > 25` → trend, directional trades allowed. `ADX < 20` → **FLAT** (stand aside). 20–25 = no-new-entry buffer.	ADX/DMI ✅
Direction	`+DI > −DI` → bullish (long side); `−DI > +DI` → bearish (short side)	ADX/DMI ✅
Confirm	SMA50 slope: `SMA50[today] −`	OHLCV ✅

Step	Rule	Data (have it?)
	`SMA50[5d ago] > 0` = up-regime, `< 0` = down-regime	
Volatility bucket	`ATR%(14) = ATR/Close` vs its ~100-day average → Low / Mid / High (sizing input)	OHLCV ✅

Decision: `allowed_side = f(ADX gate, +DI vs −DI, SMA50 slope)`.- Down-regime (-DI>+DI **and** SMA50 sloping down, ADX>25) → **SHORT_ONLY** — *longs disabled.* This is exactly what would have stopped today's bleed. - Up-regime → LONG_ONLY. Mixed/weak → FLAT.

> **The ADX gate is non-negotiable.** The +DI/-DI direction rule only survived 2-1 in verification and is repeatedly called "a recipe for disaster in a ranging market" *without* the ADX filter. DI gives direction; **ADX gives permission.** Skipping the gate reproduces today's shorting-risers / longing-fallers failure.

Sources: Fidelity/Wilder (ADX>25 trend, <20 no-trend; DI relationship), QuantMonitor (SMA50-slope + ATR% buckets).

Layer 2 — Intraday entry / flip state machine

Runs on intraday bars, **constrained by Layer 1's `allowed_side`**. States: `FLAT → LONG → SHORT → FLAT`.

- **Engine: Supertrend (ATR-based) as stop-and-reverse.** Two native states (long/short); flips when price *closes* on the opposite side of its active ATR band. The band is path-dependent — ratchets up in uptrends, down in downtrends — so it doubles as a **trailing stop** and resists premature flips. *(Source: FXEmpire + canonical TradingView formula.)*
- **Entry trigger:** `+DI crosses above −DI` (long), gated by `ADX>25`. For shorts, drive off the **-DI>+DI relationship**, *not* a mirror "-DI crosses below +DI" event — that mirror phrasing was **refuted 0-3**.
- **FLAT overlay (TradePilot's addition):** even if Supertrend signals a side, force FLAT when Layer 1 revokes permission (ADX<20, or the side isn't allowed).
- **Chop protection (the core anti-whipsaw):** stand aside when `ADX<20` **or** the Supertrend line stays flat >~5 candles while price keeps crisscrossing it. Supertrend whipsaws (4–6 flips, ~45-55% win) in range-bound tape — gating fixes it. *(Verified 3-0.)*

Flip rule, plain language

> Be on the side Supertrend says, **only if** Layer 1 allows that side **and** ADX>25. The moment Supertrend closes across its band → reverse (if the new side is allowed) or go FLAT (if not). If ADX drops below 20 → FLAT regardless.

Regime-type detection (trend vs mean-revert vs chop)

Use the **Hurst exponent (H)** as a *soft* per-horizon hint: `- H > 0.5` → trending/persistent → use the Supertrend+DMI trend logic above. `- H < 0.5` → mean-reverting → use mean-revert logic (e.g. fade extremes to VWAP). `- H ≈ 0.5` → random walk → prefer FLAT.

Caveats (verified): Hurst is **timeframe-dependent** — compute it *separately* for daily and intraday windows. It needs **100–200+ bars**, is unstable near 0.5, overestimates on small samples (use the **DFA** estimator), and **describes the recent window, not the future** — it is a classifier, not a predictor. The claim that Hurst can *directly switch* the engine was **refuted 1-2**: use it only as a hint, **confirmed by ADX**, never as a standalone switch. *(Source: Macrosynergy + peer-reviewed scale-dependence literature.)*

How this kills today's failure mode

Today's mistake	What stops it here
`LONG_IN_BEAR` (78×)	Layer 1 sets SHORT_ONLY in a down-regime → longs disabled
`SHORTED_RISER` (22×)	ADX gate + only short when -DI>+DI; intraday rule: never short a stock green/above VWAP
Trading in chop	ADX<20 → FLAT; Supertrend flat-line stand-aside
No SELL signal exists	Layer 1 produces an explicit SHORT side — the missing tier

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Validation / backtesting (strongest-sourced section — primary papers)

1. **Walk-Forward Optimization (minimum bar):** optimize on in-sample window → validate **only** on the next out-of-sample window → slide forward → aggregate **only OOS** results. A plain chronological split or plain walk-forward is **insufficient** (walk-forward ranked weakest for overfitting control).
2. **Report the Deflated Sharpe Ratio (DSR):** discounts the selection bias from testing many strategy variants (the more variants you try, the higher a Sharpe you'll hit by chance). *(Bailey & López de Prado, primary.)*
3. **Strict point-in-time data** — no look-ahead bias; only use info available at the historical bar.
4. **Tune ALL defaults via WFO** — the ADX thresholds, 5-day SMA lookback, 100-day ATR baseline, and Supertrend ATR-multiplier are arbitrary textbook defaults, **not** validated numbers. Do not ship them as-is.

5. *Optional upgrade:* Combinatorial Purged CV (CPCV) ranks best on overfitting in a 2024 Elsevier study, but the panel **declined to mandate it** (heavier to implement). WFO+DSR first.

Do NOT hardwire (explicitly refuted this run)

- **FII/DII** directional-sign, magnitude thresholds (±₹1k/2-5k/5k Cr), and 3-week-selling-streak rules → all refuted (1-2 / 0-3). Keep FII/DII as *exploratory features only*, pending your own backtest. Do not gate the regime on them.
- Mirror "-DI crosses below +DI = short" crossover → refuted 0-3.
- "Backtest length scaled to holding period" (3-4yr intraday etc.) → refuted 0-3.
- Specific "0% long activation in down regime" number → refuted (use as a *design rule* to disable, not a cited stat).

Open gaps (no verified claim covered these)

- **VWAP, RSI, MACD, India VIX, breadth** roles are unvalidated by this research. VWAP is a natural intraday flip/mean-revert anchor (and the audit already flagged "short only below VWAP"); India VIX is a natural volatility-regime input. Treat as design choices to validate, not evidence-backed.

Suggested build order

1. Layer 1 gate (ADX/DMI + SMA50 slope) → emit `allowed_side` per stock. **Wire into the scorer so it finally produces a SHORT tier.**
2. Backtest Layer 1 alone (WFO + DSR) — does the regime gate alone turn bear days green? (Today says yes.)
3. Add Layer 2 Supertrend flip machine, constrained by Layer 1.
4. Add Hurst as a soft mode-hint; add VWAP intraday rule.
5. Paper-trade a new engine (`v7_regime`?) alongside v4/v5 for A/B.